

2 - Exploring the model

October 8, 2024

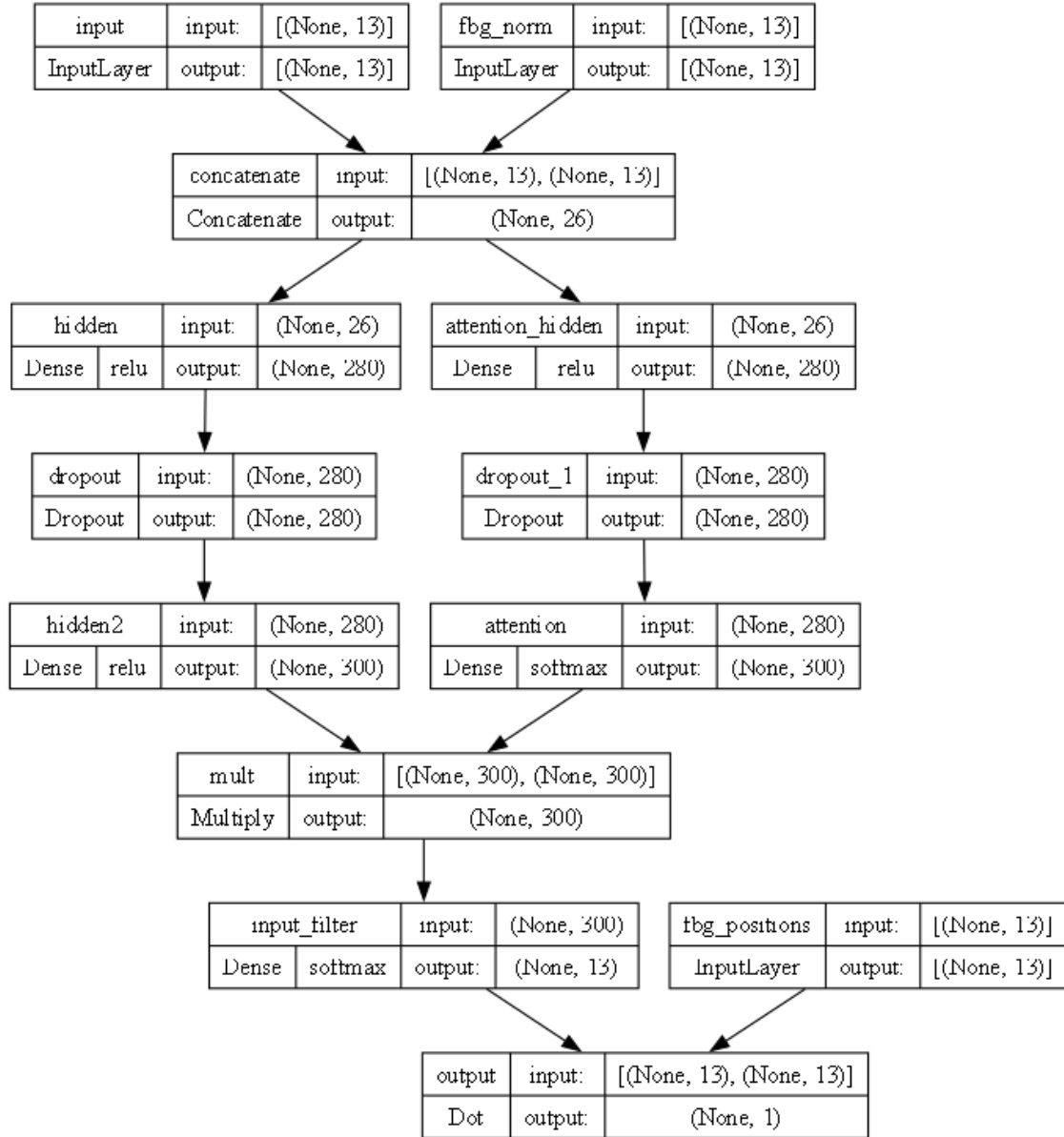
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WARNING:tensorflow:From C:\Users\felip\AppData\Roaming\Python\Python310\site-packages\keras\src\losses.py:2976: The name tf.losses.sparse_softmax_cross_entropy is deprecated. Please use tf.compat.v1.losses.sparse_softmax_cross_entropy instead.
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1 Load the model

Load the LPFG decoder model and the fine tuned weights.

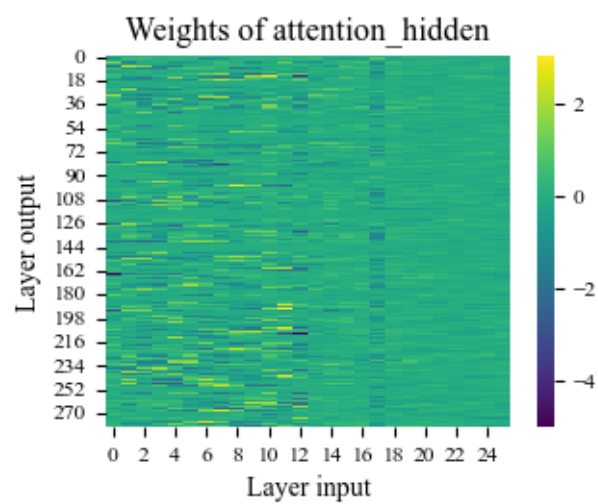
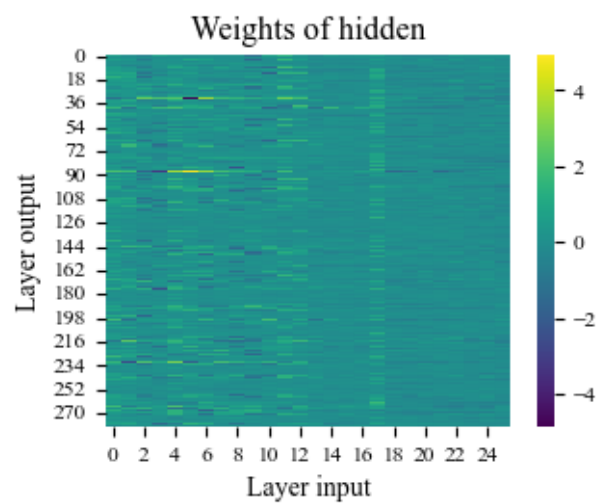
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WARNING:tensorflow:From C:\Users\felip\AppData\Roaming\Python\Python310\site-packages\keras\src\backend.py:1398: The name tf.executing_eagerly_outside_functions is deprecated. Please use tf.compat.v1.executing_eagerly_outside_functions instead.
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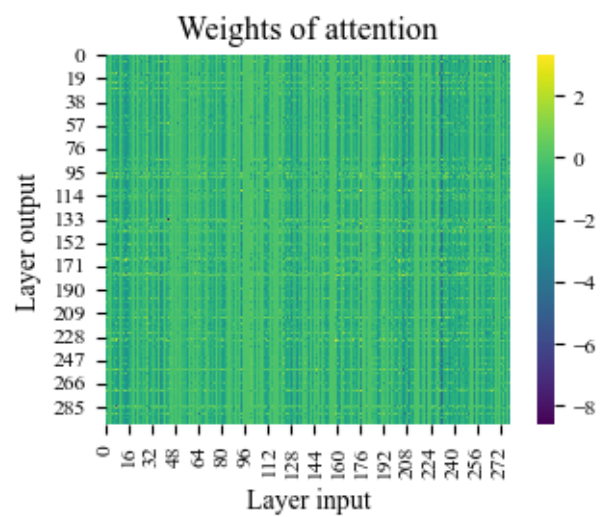
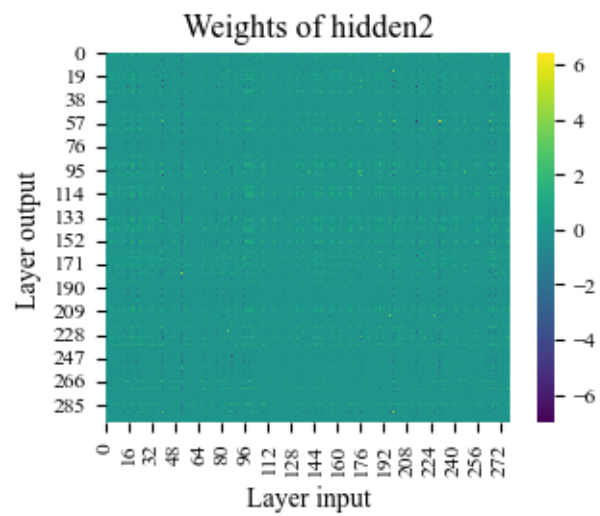
Below one can see the model architecture.

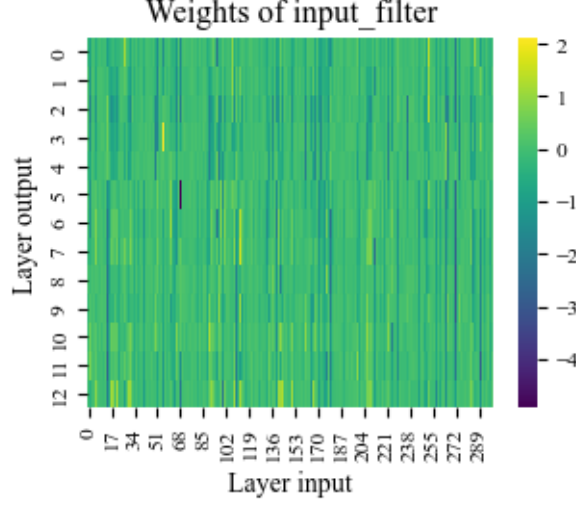


2 Model weights

Display model weights. Below one can see the weight connecting each layer, the x-axis represents the input layer number and the y-axis the output layer.







3 Layer output as function of LPFG parameters

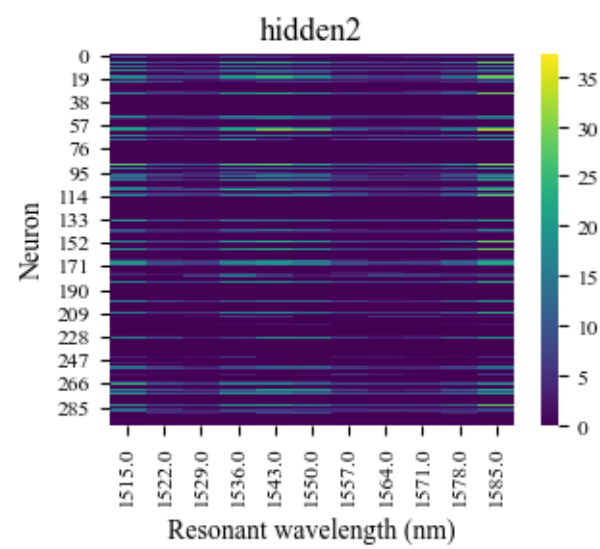
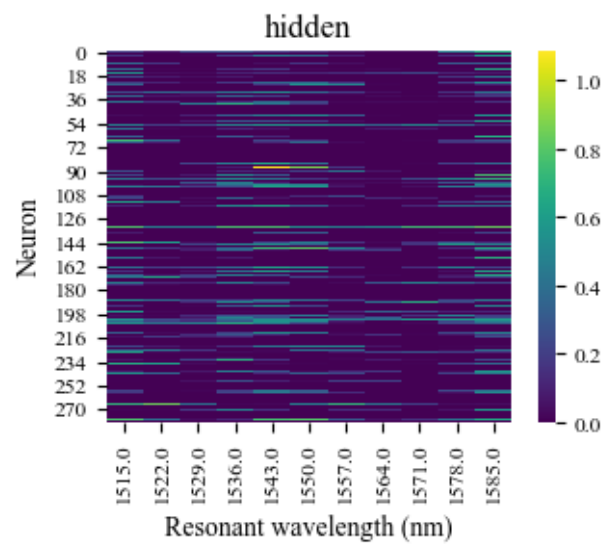
Below we show what happens inside the model, i.e. each layer output for a given input.

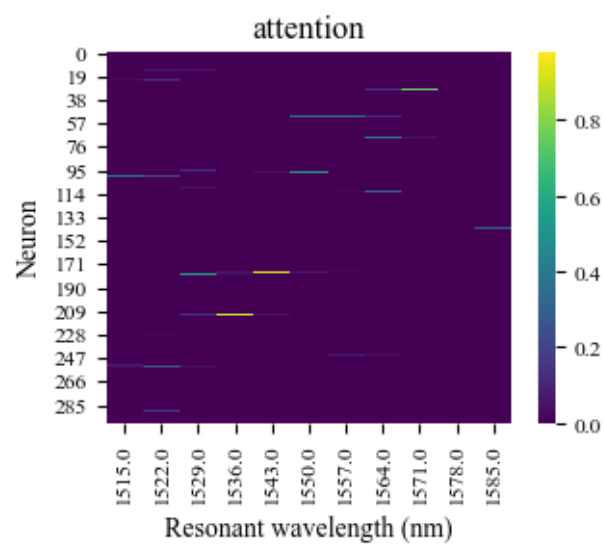
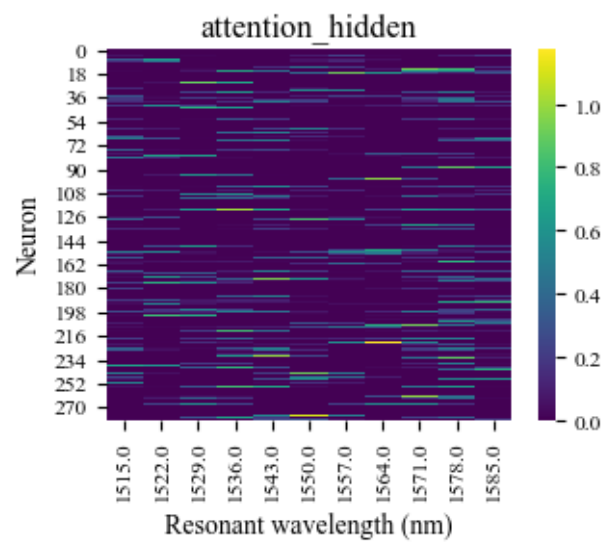
For better control, synthetic data was considered and only one LPFG parameter changed at time. And only LPFG spectra without noise was considered. On the next section the attention mechanism will be further explored and the impact of noisy LPFG analyzed.

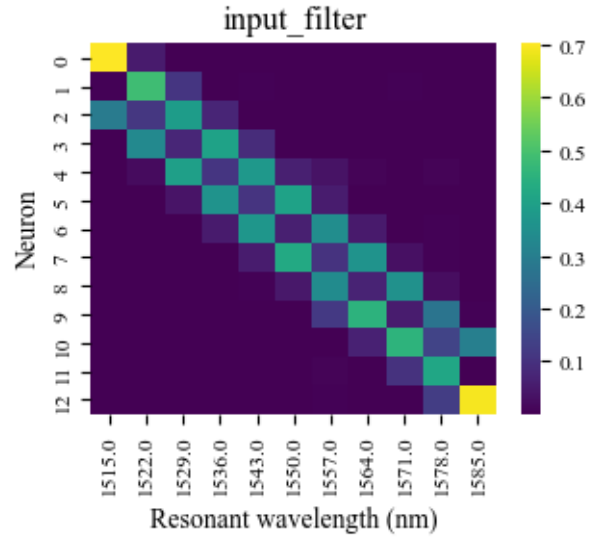
Let us consider the following parameters: Resonant wavelength, the Resonant dip depth, and the Resonant dip width.

3.1 Resonant wavelength

Let us analyse the impact of the resonant wavelength. Hence, one can see which neurons are activated by each spectral region of LPFG spectra.

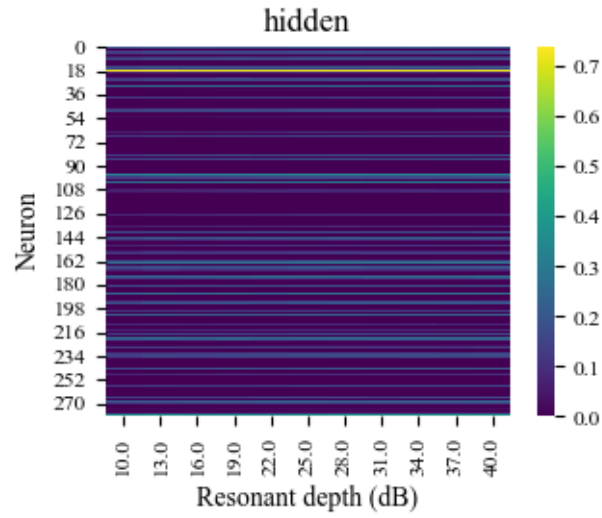


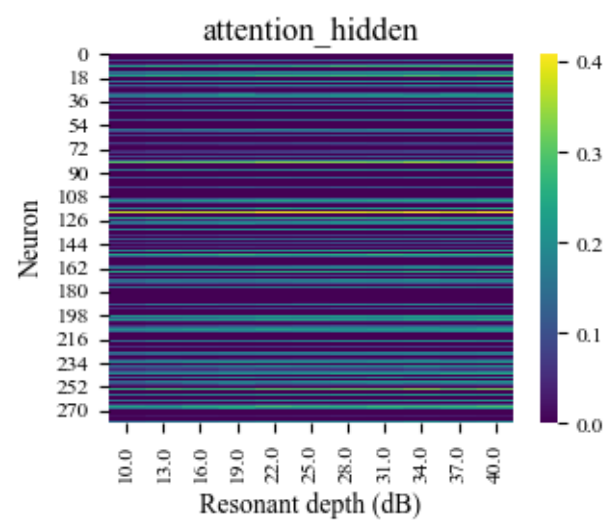
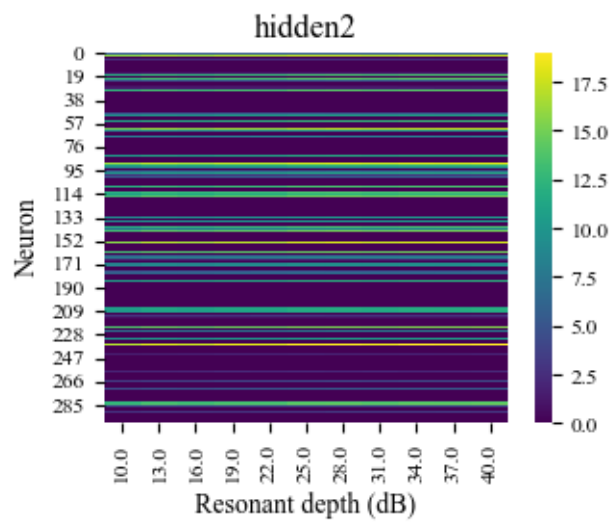


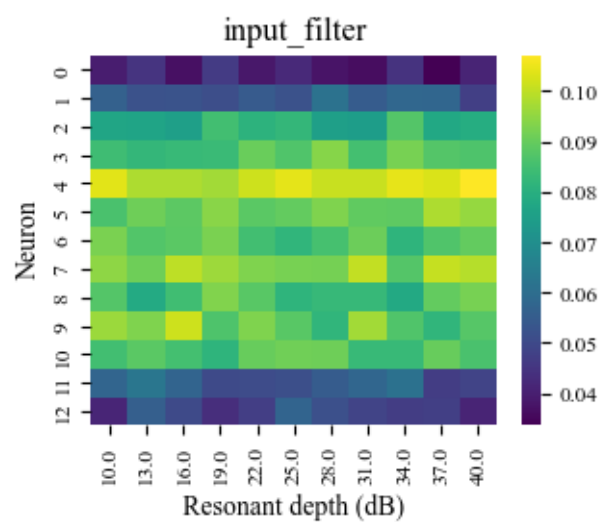
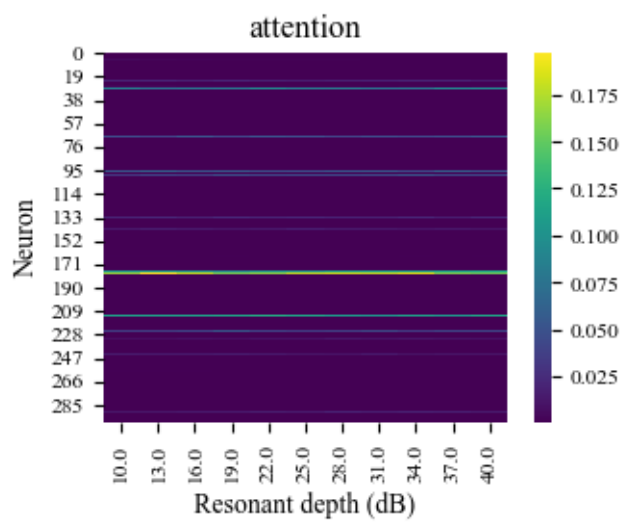


3.2 Coupling efficiency

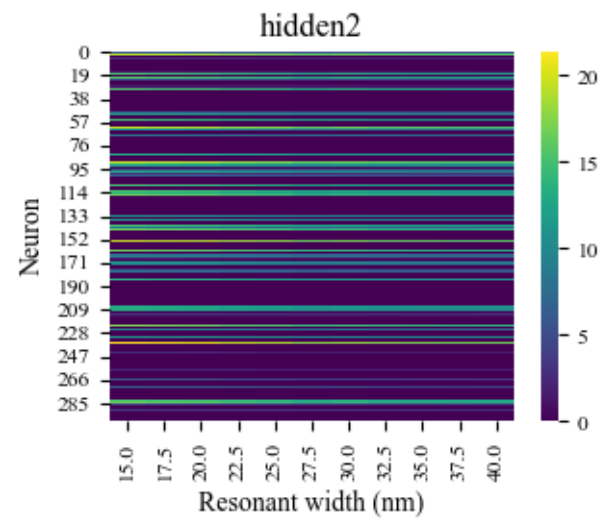
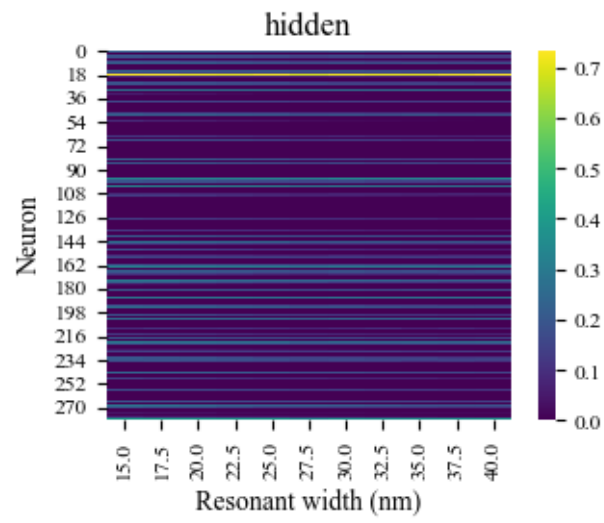
Let us analyse the impact of coupling efficiency in the form of the resonant dip intensity.

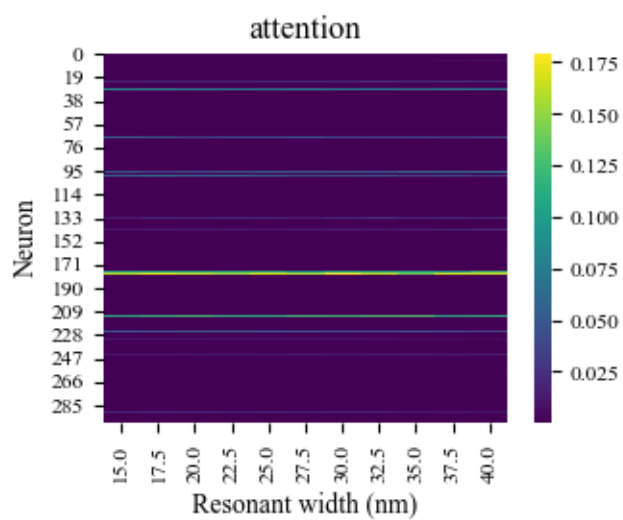
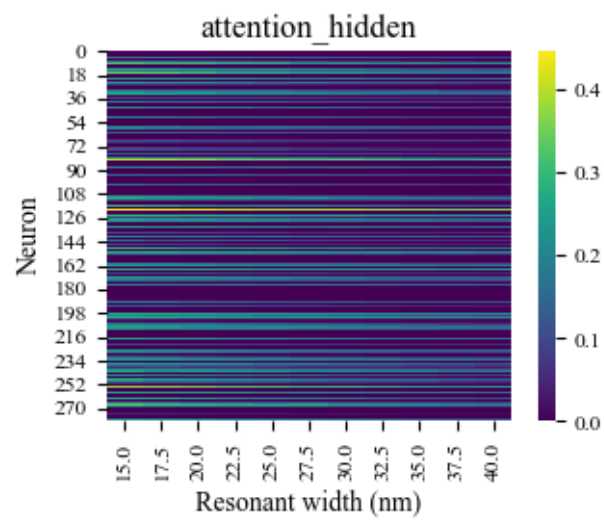


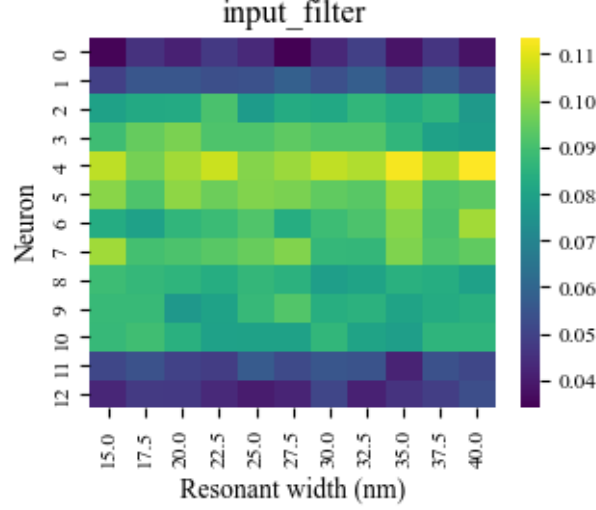




3.3 FWHM







3.4 Conclusions from neuron activation

Considering the results obtained varying several LPFG parameters and observing the activation of each neuron of the network, one could note that the neuron activation is highly sensitive to the LPFG resonant wavelength. Hence, the weights encode the resonant wavelength, with a singular neuron combination for each spectral region.

For the dip depth and width, there was no clear correlation with the neuron activation. These results showed the model effectively focus on the resonant wavelength information because the mean neurons' behavior was only regarding the resonant wavelength.

4 Exploring attention

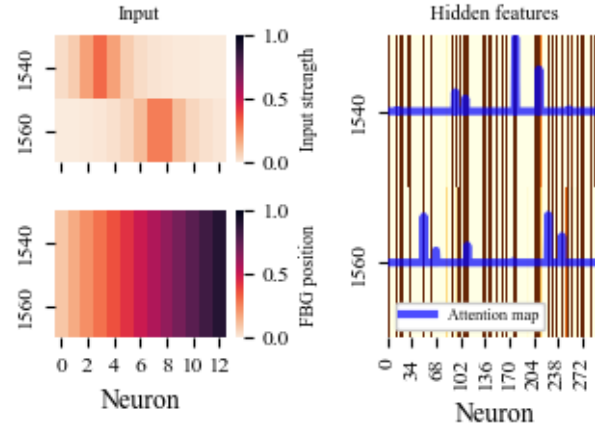
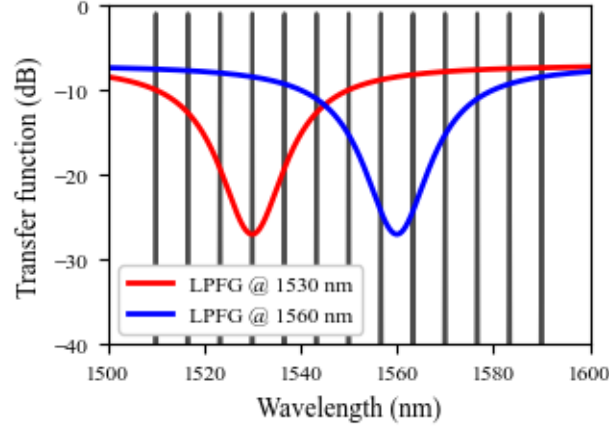
Below we explore the attention mechanism, showing how the system behaves given two different LPFGs.

4.1 LPFG position effect

Here we show how the outputs change given a LPFG shift. It is the same analysis as presented in the *Layer outputs by resonant wavelength* section, but with a synthetic spectrum, static FBG array and input demonstration, for better visualization.

1540: 1530.22 +- 0.12

1560: 1560.01 +- 0.07



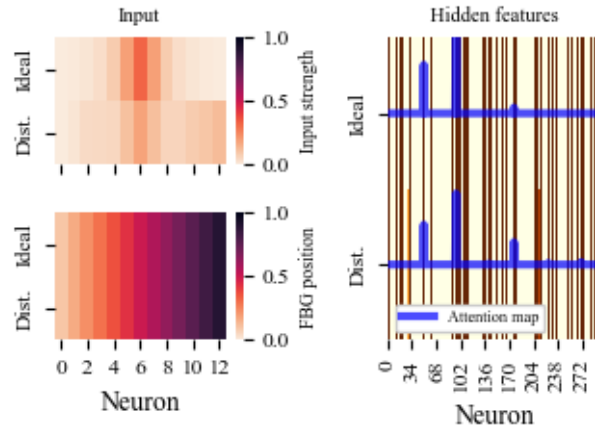
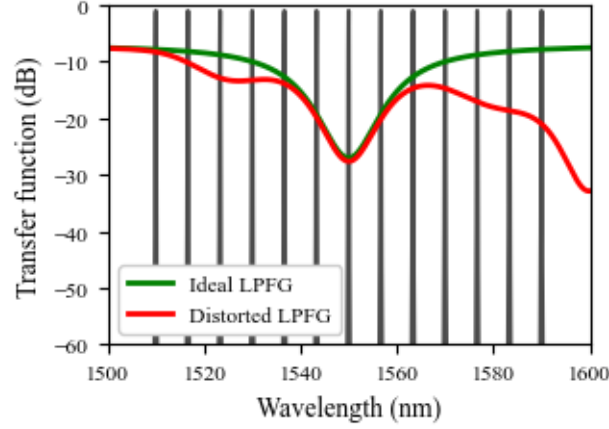
4.2 Ideal vs distorted LPFG

Below we create a LPFG transmission function with a single distinct resonant dip and another one with some distortions and a second dip outside the interrogator range.

This might be the most influential part of attention!

Ideal: 1549.86 +- 0.09

Distorted: 1549.91 +- 0.19



4.3 Designed FBG array vs shifted FBG array

We explored the effect of FBG shift, by considering the FBGs at the designed Bragg wavelength and at arbitrary shifted positions.

One can see a great difference at the input layer, but a very good estimation for both. The attention and second hidden effects are similar to the shown before.

Ideal: 1549.88 $\pm$ 0.06

Distorted: 1549.78 $\pm$ 0.25

